

else Statement

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Java

We know how to ask a question with an if statement

```
boolean answer = x < y;  
if(answer){  
    System.out.println(x + " is smaller than y!");  
}
```

This runs if the statement is true,
but what if it's not???

```
boolean answer = x < y;  
if(answer){  
    System.out.println(x + " is smaller than y!");  
}
```

The else statement

The else statement is a catch all for all FALSE conditions.

This MUST be paired with an if statement and is only tied to that branch.

```
if( /* Condition */ ){  
  
}  
else{  
  
}
```

The else statement

Notice: The else statement does not have a condition.

Else statements run when the if statement and all conditions above are false.

```
if( /* Condition */ ){  
  
}  
else{  
  
}
```

The else statement example

```
if(5 < 3){  
    System.out.print("This doesn't run.");  
}  
else{  
    System.out.print("This does run.")  
}
```

But else statements are catch alls!

```
if(x < 10){  
    System.out.print("x is less than 10");  
}  
else{  
    System.out.print("What am I?");  
}
```

But else statements are catch alls!

```
if(x < 10){  
    System.out.print("x is less than 10");  
}  
else{  
    System.out.print("What am I?");  
}
```

The else runs when x is **greater than or equal to 10!**
For whatever value x may be!

Branches

This is a singular branch.

It **can** have one outcome and only one.

```
if(){  
  
}
```

Branches

These are two separate branches.

They both **can** run, but don't have to run.



```
if(){
```

```
}
```

```
if(){
```

```
}
```

Branches

This is also a **singular branch**.

It **can** have one outcome and only one.

They are tied together, only one can run.

```
if(){
```

```
}
```

```
else{
```

```
}
```

Branches

Though, with this singular branch there **MUST** be at least one output.

The else statement ensures that one of these statements will run.

```
if(){  
  
}  
else{  
  
}
```

else statement - Lab

Write a java program that is a number guessing game. The user must guess a number between 1 and 1000. The program will just give them one attempt to guess that number and tells the user whether the user was correct or not. Make sure to print out that number at the end.